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JDBC (PART 12)

Transaction Management in JDBC

Savepoint

- Savepoint is a checkpoint in transaction and we can rollback to that perticular checkpoint
- Savepoint is automatically released/invalid when we use methods i.e. commit() or rollback()
- Savepoint is an Interface which was introduced in JDBC 3.0 version
- ts implementation is provided by database vendors
- Syntax :

```
package java.sql;

interface Savepoint

{

    getSavepointId()
```

```
        getSavepointName()
}
```

- Some savepoint methods are provided by Connection interface
  - con.setSavePoint() - Savepoint
  - con.setSavePoint(String name) - Savepoint
  - con.releaseSavepoint(Savepoint sp) - void

- Many databases supports savepoint but some doesnt
- We can check that any database supports savepoint or not by DatabaseMetaData

Isolation : All the transactions must be executed independently, one transaction must not give effect to another transaction

- Transaction Consistency Problem :-

  1. Lost Update
  2. Dirty Read
  3. Non-Repeatable Read
  4. Phantom Read

- To remove above problems Connection interface provides 5 isolation levels

  1. TRANSACTION_NONE or 0 (It indicates that transaction is supported or not)

2. RANSACTION_READ_UNCOMMITTED or 1

3. TRANSACTION_READ_COMMITTED or 2

4. TRANSACTION_REPEATABLE_READ or 4

5. TRANSACTION_SERIALIZABLE or 8

- Connection interface provides 2 methods to know isolation levels

  1. getTransactionIsolation()

  2. setTransactionIsolation(int level)

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2. TRANSACTION_READ_UNCOMMITTED or 1

- It is lowest isolation level

- It only solves Lost Update Problem
- This isolation level is not recommended

3. TRANSACTION_READ_COMMITTED or 2

- This isolation level ensures that only committed data can be read by other transactions
- It only solves Lost Update & Dirty Read Problem

4. TRANSACTION_REPEATABLE_READ or 4

- Mostly databases has by default this isolation level (MySQL)

- It solves Lost Update, Dirty Read & Non-Repeatable Read problem

5. TRANSACTION_SERIALIZABLE or 8

- This is highest isolation level
- This isolation level is not recommended because it can create performance problem
- It can solve Lost Update, Dirty Read, Non-Repeatable Read & Phantom Read

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